

■ MandiQ Project Architecture & Working Guide

Detailed Explanation in Hinglish (Vercel, Render, Database, Frontend & Backend ML Models)

1. MandiQ Overview (Introduction)

MandiQ ek AI-powered Mandi Price Prediction & Analytics Application hai. Iska main target farmers aur traders ko market information dena hai, taaki ve apni crop ko sahi daam aur sahi mandi me bech sakein. Is application me daily price tracking, next 7-30 days price forecast, multiple mandis ka dynamic comparison (after transport cost deduction), aur custom price alerts ke features shaamil hain.

2. Frontend Architecture & Screens

Frontend ko **React + Vite + TypeScript + TailwindCSS** ka use karke banaya gaya hai. Mobile app builds ke liye isme **Capacitor** configure kiya gaya hai, jo direct Android APK/iOS app compilation me help karta hai. Frontend routing ke liye React Router (react-router v7) ka use kiya gaya hai.

App me total 11 Screens aur ek Onboarding Wizard hai:

A. Public / Auth Screens — Inhe bina login ke access kiya ja sakta hai.

- **SplashScreen**: App launch hote hi check karta hai ki user logged-in hai ya nahi aur respective screen par send karta hai.
- **LoginScreen**: Yahan user apna 10-digit mobile number enter karke OTP verify karne ke liye submit karta hai.
- **OtpScreen**: 6-digit OTP code enter karne ke liye. Developed features ko test karne ke liye '000000' master bypass code lagaya gaya hai.

B. Onboarding Wizard — New users ke login karne ke baad setup wizard chalta hai:

- **UserTypeScreen**: Role select karne ke liye (Farmer / Trader).
- **NameScreen**: User ka basic name input karne ke liye.
- **LocationScreen**: State, District aur Village setup karne ke liye.
- **CropSelectionScreen**: Track karne wali crop (Tomato, Potato, Onion, Spinach) choose karne ke liye.

C. Protected Main App Screens — Login aur onboarding ke baad accessed screens (Bottom Navigation ke through):

- **HomeScreen**: Dashboard jahan Mandi aur Crop select karne ke baad 7-day strip (Actual & Predicted hybrid list), price trends chart aur best mandi compare features dikhte hain.
- **CropBrowseScreen**: Track hone wali sabhi crops ka overview list.

- **PredictionScreen:** 7-day forecasting chart, upper/lower pricing band, prediction confidence level percentage me, aur dynamic AI advice card (Sell/Hold) show karta hai. Feedback submit karne ka input field bhi yahan hai.
- **MandiCompareScreen:** Sabhi available mandis ke transport costs ko real price se minus karke net profit clear compare karta hai.
- **MandiInfoScreen:** Mandi locations, coordinates, address aur helpline numbers details ke liye.
- **AlertsScreen:** Custom Price Alerts (Above/Below target price) select aur create karne ke liye option deta hai.
- **PastTrendScreen:** Historical graphs, monthly averages, yearly trends aur seasonal parameters (best months to buy or sell) dikhata hai.
- **ProfileScreen:** Profile details aur languages (English, Hindi, Punjabi, Marathi) manage karne ke liye.

Assistant Chatbot Widget (SupportChat):

HomeScreen aur baaki screens par ek floating chat bubble hai jo multilingual query solver hai. Isme keyword dictionary mapping algorithm hai jo text ya voice input me se relevant keywords (jaise 'free', 'price', 'mandi') detect karke 4 native languages (English, Hindi, Punjabi, Marathi) me response deta hai.

3. Backend Architecture & API Flow

Backend ko **FastAPI (Python)** framework me design kiya gaya hai. Iska structural workflow highly asynchronous hai. Sabhi heavy processes jaise database updates ya model training background tasks ke through execute hote hain.

Main Components:

- **FastAPI Application (main.py):** Sabhi REST API endpoints handle karta hai. Isme strict CORS policy set hai.
- **Daily Scraper & Captcha Solver:** Agmarknet portal se live data fetch karne ke liye daily scheduler set hai. Agmarknet APIs me raw captcha request check hoti hai. Isko solve karne ke liye backend **2Captcha API** use karta hai. `fetch_captcha()` function image download karta hai, 2Captcha use resolve karke answer code return karta hai, tabhi daily data pull hota hai.
- **APScheduler Cron Jobs:** `server.py` lifespan me BackgroundScheduler chalta hai:
 - Daily 6:00 AM IST: Sirf aaj ka new data pull/scrape karke database me insert karne ke liye.
 - Sunday 2:00 AM IST: Full retrain `run_pipeline.py` run karta hai sabhi commodities ke liye.
 - Daily 9:00 AM & 6:00 PM IST: Alerts validation check karta hai aur SMS status check change karta hai.
- **Parser Services:** `pdf_parser.py` aur `csv_parser.py` upload kiye gaye agmarknet records (PDF aur CSV files) ko parse karke structured rows database format me clean karte hain.

4. Database & Authentication

Database Setup: MandiQ me hybrid setup hai. `database.py` file automatic routing check karti hai. Agar server me environment variable **MONGO_URI** set hai, to saari operations direct cloud MongoDB Atlas me routing ke through save hoti hain. Agar cloud MONGO_URI set nahi hai (local setup me), to local directory database **SQLite3 (data/mandiq.db)** automatic fallback target banta hai.

Authentication Flow:

1. User mobile number enter karta hai, backend random 6-digit OTP create karke 5 min expiry time ke sath database tables (otps collection) me save kar leta hai.
2. OTP SMS ko user ke pass send karne ke liye **Renflair SMS Gateway API** requests use ki jati hain.
3. OTP verify hone par user login successful hota hai. Ek user session **JWT Token (30 days validity)** generate kiya jata hai. JWT token ko local storage me persist kiya jata hai aur har authenticated route me authorize bearer header ke through cross-validate kiya jata hai.

5. Machine Learning Models & Forecasting

MandiQ me pricing forecast ke liye 4 different regressors ka weighted ensemble algorithm run kiya jata hai. Isme hum features extract karte hain jo models ko correct patterns learn karne me help karte hain.

Ensemble Models & Weights:

- **Random Forest Regressor** (40% Weightage) — Noise aur fluctuations handle karta hai.
- **Extra Trees Regressor** (40% Weightage) — Generalization boost aur variance drop karta hai.

- **XGBoost Regressor** (10% Weightage) — Bias matching me help karta hai.
- **LightGBM Regressor** (10% Weightage) — Fast gradient boosting calculations ke liye.

Feature Engineering (69 Total Features):

- **Calendar Features:** Month, day of year, cyclic sin/cos calendar mappings, Indian crop seasons.
- **Autoregressive Lags:** Past 1, 2, 3, 5, 7, 14, 21, 30 days actual prices.
- **Rolling Windows:** 7, 14, 30, 60 days rolling average (mean) aur standard deviation (volatility).
- **Supply Dynamics:** Crop arrival quantities (MT) at mandis, log volumes, price-to-arrival ratios.
- **Meteorological Inputs:** Local weather (temperature min/max, rainfall, humidity) delhi and source region of the crops dynamically shifted by crop-specific harvest delay lags.

Multi-step Recursive Forecasting:

Hamara model recursively work karta hai. Kal ka price forecast karke, use next day ke input features me update kar diya jata hai. Ensemble predictions ka standard deviation variance check deta hai jise upper aur lower limit price bands set karne me use karte hain.

6. Production Hosting & Deployment

A. Frontend Deployment (Vercel):

Frontend Vercel platform par hosted hai. Yahan *vercel.json* rules apply hain jo all client side URLs ko rewrites karke */index.html* par hit karte hain. Isse browser refresh karne par 404 error nahi aati.

B. Backend Deployment (Render):

Backend Render platform par Python Web Service target structure me deploy hota hai. Ye config *render.yaml* ke true constraints se dynamic set kiya gaya hai. Build commands dependencies download karti hai aur startup me uvicorn execution run hota hai.

C. Database Setup:

MongoDB Atlas cloud connection coordinate dynamic settings variable (MONGO_URI) control se connect hoti hai. Isse scaling or live multi-user concurrency highly manageable aur secure rahti hai.